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Topic: Energy Generation Using Vortex-Induced Vibrations in the Zhambyl
Region: the Ventrix Bladeless Wind Generator

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Отзыв

Работа является глубоким и самостоятельным исследованием, выполненным на стыке физики, экологии и региональной экономики. Актуальность темы не вызывает сомнений, так как проблема энергодефицита и поиска альтернативных источников энергии для нашего региона является крайне важной.

Отличительной чертой данной работы является ее практико-ориентированный характер. Учащийся не ограничился теоретическим анализом, а провел полный цикл исследования: от изучения фундаментального физического явления (вихревые колебания) до создания работающего прототипа, способного генерировать электроэнергию. Это демонстрирует высокий уровень понимания предмета, инженерный подход и прекрасные экспериментальные навыки.

Все этапы работы выполнены последовательно и грамотно. Проект структурирован, гипотеза четко сформулирована и доказана. Особо хочу отметить качественный анализ преимуществ и недостатков технологии, а также расчет ее потенциальной эффективности именно для условий Кордайского района, что говорит о глубоком погружении автора в проблему.

Данная работа является образцовой и заслуживает самой высокой оценки. Она может быть рекомендована для участия в республиканских и международных научных конкурсах проектов.

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Жайк

Отзыв

Работа Ошан Дианы, представленная на рассмотрение, вызывает несомненный научный и практический интерес. Автор взялся за сложную и современную задачу, находящуюся в фокусе внимания мировой науки – поиск новых, эффективных и малозатратных технологий в области ветроэнергетики.

Выбор темы, основанный на анализе региональных проблем энергоснабжения, является грамотным и обоснованным. Методология исследования выстроена корректно: от теоретического обзора фундаментальных основ механики жидкости и газа (явления вихреобразования и автоколебаний) до проектирования, изготовления и испытаний экспериментальной установки. Получение конкретного результата в виде измеренной мощности генерации (порядка 97 Вт) является убедительным доказательством состоятельности предложенного технического решения.

Особую ценность работе придает проведенный сравнительный анализ с традиционными установками и экономическое обоснование, что редко встречается в проектах школьного уровня и свидетельствует о системном мышлении автора.

Несмотря на школьный уровень, работа соответствует формату серьезного инженерного исследования. Предложенные автором рекомендации по дальнейшей оптимизации конструкции (повышение КПД) являются логичным продолжением работы и могут быть положены в основу будущих изысканий.

Данное исследование заслуживает высшей оценки и безусловной победы на конкурсах научных проектов. Автор проявил выдающиеся способности в области физики и технического творчества и демонстрирует высокий потенциал для дальнейшего обучения и научной деятельности в ведущих технических вузах страны.

Бижигитов профессор



Abstract

Relevance: The Zhambyl region, particularly the Kordai district, faces an energy deficit and dependence on external supplies. Implementing bladeless wind generators based on vortex-induced vibrations can provide a stable and environmentally friendly energy source.

Objective: To address the energy deficit in the Zhambyl region by studying the potential of the bladeless Ventrix system and developing an autonomous energy technology.

Hypothesis: The proposed technology can mitigate the energy deficit in the Kordai region, improving living conditions, supporting infrastructure development, and reducing fossil fuel use.

Research stages:

Task 1. Justify wind energy as a solution and identify the limitations of traditional turbines.

Task 2. Study vortex formation and develop a vibration-to-electricity conversion mechanism.

Task 3. Assemble and test the Ventrix prototype to determine operational parameters.

Task 4. Improve the prototype through design modification and amplitude control implementation.

Methods: Theoretical analysis of vortex-induced vibrations; aerodynamic analysis; evaluation of economic efficiency; climatic and comparative analysis of implementation in the Zhambyl region.

Research novelty: Ventrix is the first school-level development of a bladeless wind generator capable of stable operation under strong wind loads due to an innovative magnetic regulation system.

Conclusions and results: The research confirmed the effectiveness of Ventrix technology and vortex-induced vibrations for energy generation. The prototype demonstrated stable performance and scalability potential. Design optimization enables continuous operation and applicability for reducing energy deficit in the Kordai region.

Practical application: Ventrix can be used for autonomous power supply and for modular bladeless wind farms.

Degree of independence: The project was completed independently, with consultative support from the supervisor.

Introduction

Energy security of Kazakhstan's regions remains one of the key issues of national development. Despite the active implementation of renewable energy sources, the Zhambyl Region—particularly the Kordai District—continues to experience an electricity deficit and strong dependence on external supplies. According to *Zhambyl Electric Networks LLP*, up to 90% of the electricity consumed in the Kordai District is transmitted from Kyrgyzstan, which makes the region vulnerable to power outages and instability in energy supply.

The development of wind energy is considered one of the most promising directions, as the average annual wind speed in the Kordai District is 5–7 m/s, which is sufficient for the efficient operation of wind generators. However, conventional wind turbines have several significant drawbacks: they require large areas (up to 0.6 km² per installation), generate noise pollution of up to 80 dB(A), and pose a threat to rare bird species migrating through the Zhambyl Region. In

addition, turbine blades are subject to wear and account for approximately 29% of all equipment failures. These factors limit the large-scale deployment of traditional wind turbines.

Therefore, there is a need to search for alternative and more sustainable solutions in the field of wind energy. One such solution is the use of vortex-induced vibrations (VIV), a natural phenomenon that occurs when airflow passes around a body. Based on this effect, the bladeless wind generator *Ventrix* was developed, which converts the kinetic energy of wind into electrical energy through structural oscillations.

The aim of this study is to investigate the feasibility of using the bladeless Ventrix system to address the energy deficit of the Kordai District and to provide a theoretical and experimental justification of its efficiency.

To achieve this aim, the following objectives were set:

1. To justify the relevance of wind energy development in the Zhambyl Region and identify the shortcomings of traditional turbines.
2. To study the physical principles of vortex formation and the mechanisms of converting mechanical energy into electrical energy.
3. To design and assemble a prototype of the Ventrix generator.
4. To conduct tests and evaluate the efficiency of the prototype.
5. To propose ways to optimize the design in order to improve its stability and performance.

The study employs methods of theoretical analysis of aerodynamic characteristics, computational and experimental methods to determine the parameters of vortex-induced vibrations, as well as economic and environmental comparisons of the Ventrix system with conventional wind turbines.

Research Section

Task 1. To justify the choice of wind energy as a solution to the problem of energy deficit and to identify the shortcomings of traditional wind turbines.

1.1 The Problem of Energy Deficit

Zhambyl Region is one of the leaders in Kazakhstan in the development of renewable energy sources [1]. However, despite this progress, the issue of energy security in the Zhambyl Region—especially in the Kordai District—remains highly relevant. As noted by Almas Chukin, managing partner of the investment company *Visor Kazakhstan*, “In Kazakhstan, 70–75% of electricity is generated in the north, where the coal basin is located. However, high energy consumption in the south has forced the Kazakhstan Electricity Grid Operating Company to build three major transmission lines to transfer electricity from the north to the south. This is, of course, expensive, involves significant transmission losses, and results in unnecessary transportation costs” [2]. According to *Zhambyl Electric Networks LLP*, up to 90% of the electricity consumed in the Kordai District is transmitted from Kyrgyzstan. Such figures indicate a high level of dependence on external energy supplies, and any disruptions in neighboring territories directly lead to local energy crises [3].

Moreover, the shortage of electricity hinders the industrial development of the district. Almas Chukin also emphasizes that energy plays a key role in economic development: investors cannot build factories without a reliable electricity supply; therefore, the energy sector must develop faster than the economy itself [4]. Consequently, regional dependence on external energy sources constrains economic growth, making the search for additional and autonomous energy sources particularly urgent.

1.2 Justification for the Choice of Wind Energy and the Drawbacks of Traditional Wind Turbines

Naturally, the highly favorable climatic conditions of the Kordai District for the operation of wind power plants have attracted both investors and state support. The average annual wind speed in the district is approximately 5–7 m/s, which is sufficient for the efficient operation of wind generators, while the steppe landscape provides minimal obstacles to airflow, increasing installation efficiency. As a result, several wind power plants have been constructed in the area, generating about 200 thousand kWh per month, which covers approximately only 2.2% of the district's energy demand. However, due to limited capacity, seasonal fluctuations, and a number of significant drawbacks, traditional wind turbines are unable to fully solve the problem of energy deficit.

According to approximate practical calculations, the land area that must be allocated for a single onshore 3 MW wind turbine ranges from about 0.37 to 0.60 km², which makes their placement difficult in densely populated regions [5]. Moreover, rotating blades generate noise pollution of up to 80 dB(A), necessitating strict compliance with sanitary standards [6].

In addition, Kazakhstan currently has no restrictions on the construction of wind power plants in habitats of rare species or along migratory corridors, nor are there sufficient bird protection measures in place. The Zhambyl Region, in particular, is crossed by numerous bird migration routes. The Zhanatas Wind Power Plant has already been built on the Karatau Ridge, an area through which approximately one million birds of prey migrate during the autumn season, including between 32.3 and 40.6 thousand eagles of three species (steppe eagle, imperial eagle, and greater spotted eagle) [7]. Experience from China shows that

an increase in the number of turbines by one standard deviation reduces bird populations by 9.75% and leads to a 12.2% decrease in bird species richness at the county level [8]. Thus, the use of traditional wind generators in the Zhambyl Region may reduce ecosystem biodiversity and put rare eagle species and other birds of prey at risk of extinction.

Furthermore, the International Institute for Energy Transition Analysis (IETA) has found that the component of traditional wind turbines subject to the greatest wear is the blades, which account for approximately 29% of all equipment failures. These components most frequently require maintenance or replacement [9]. This situation highlights the relatively low efficiency of conventional green energy generators and underscores the need to identify alternative solutions for regional development and improved quality of life for residents.

One of the most promising directions for the development of wind energy in the region—and a potentially optimal solution to the energy deficit in the Kordai District—is the use of vortex-induced vibrations (Vortex-Induced Vibrations, VIV) as an alternative method of electricity generation. This mechanism is based on the natural phenomenon of vortex formation when airflow passes around a body, causing periodic oscillations of the structure. This study examines the bladeless wind generator *Ventrix*, which utilizes this principle to convert wind energy into electrical energy.

Unlike traditional turbines, the *Ventrix* system has fewer moving parts, which reduces wear, lowers noise levels, and simplifies maintenance. Given the region's clear potential for wind energy development, this technology may serve as an effective solution to the energy deficit, opening new opportunities for improved living conditions of local residents and sustainable industrial development of the region.

Task 2. To investigate the physical basis of vortex formation and to develop a mechanism for converting oscillations into electrical energy.

2.1 Resonance

One of the most well-known cases of aerodynamic resonance is the collapse of the Tacoma Narrows Bridge. The phenomenon of resonance occurs when oscillations are amplified by a periodic external force. In aeroelasticity, airflow can induce oscillatory motion of a body if its natural resonant frequency is close to the frequency of vortex shedding in the flow. Oscillations induced by vortices are known as the VIV phenomenon (vortex-induced vibrations) (Fig. 1) (Appendix 1).

The Principle of Electrical Energy Generation in a Vortex-Induced Vibration System

The generation of electrical energy in this system is based on the mechanical transfer of energy from vortex-induced vibrations to a stepper motor. The structure (Fig. 2) consists of a cylindrical mast with an internal axis that divides the system into two parts: an outer section L1 and an inner section L2. Vortex flows, formed as air passes around the structure, create transverse oscillations that set L1 in motion. This motion is transmitted to L2, which is located inside the casing and connected to the energy transmission system.

Mechanism of energy transfer from vortex oscillations

Vortex shedding occurs as a result of the periodic detachment of vortices from the surface of a cylindrical structure, which leads to transverse forces acting perpendicular to the airflow direction. This process can be described using the Strouhal number (St) (typically in the range of 0.15–0.2 for cylinders), which relates the vortex shedding frequency f to the characteristic length D of

the structure and the airflow velocity V .

The oscillations of the external part L_1 are induced by these aerodynamic forces and are transmitted to the internal part L_2 , which is fixed inside the

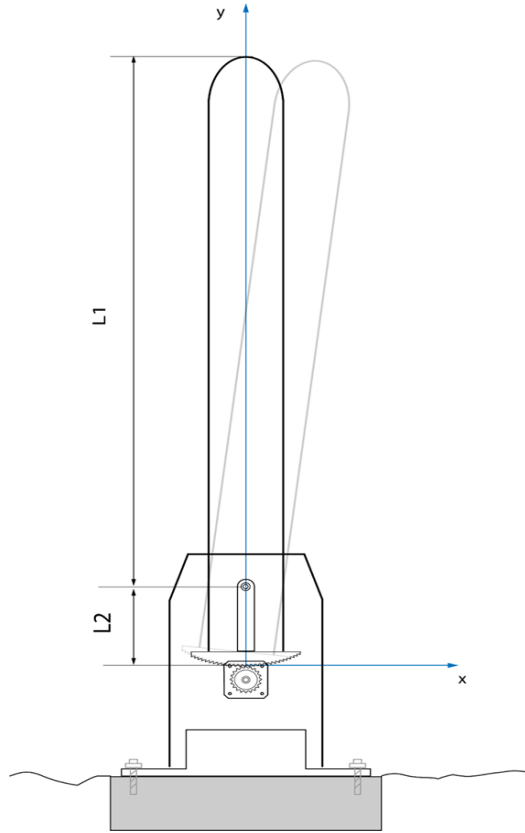


Fig. 2. Ventrix Wind Turbine Design

2.2 Transmission of Mechanical Energy to the Generator

The oscillatory system includes a gear transmission connected to L_2 . At the lower part of the structure, a gear wheel is installed, which interacts with a rack-and-pinion mechanism connected to the rotor of a stepper motor. This mechanism allows the conversion of reciprocating motion into rotational motion, ensuring efficient transmission of mechanical energy into electrical energy.

Using a stepper motor instead of traditional electromagnetic induction-based generators provides several advantages. One of them is increased accuracy in

housing.

The difference in length between L_1 and L_2 causes the internal part to experience oscillations with higher frequency and amplitude. This effect is amplified by the mechanical coupling between the structural elements, allowing the accumulated energy to transfer from the external part to the internal part, increasing the vibration intensity.

converting mechanical oscillations into an electrical signal due to the discrete operation of the stepper motor. In addition, such a system is stable under variable wind speeds thanks to controlled rotor rotation, which reduces energy losses and increases generation efficiency. Another important aspect is the ability to operate at low airflow speeds, making the system effective even under weak wind conditions and expanding its applicability across different climatic environments.

2.3 Resonance Phenomena and Structural Optimization

One of the key factors affecting the system's efficiency is tuning the structure to the resonance frequencies of vortex shedding. Vortex-induced oscillations are amplified when their frequency matches the natural frequency of the structure. The natural frequency of the oscillatory system is determined by the stiffness k and mass m :

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

Where k is the stiffness coefficient of the structure, and m is the mass of the oscillating element. For efficient energy harvesting, the system should be tuned so that the natural frequency of the internal part L_2 falls within the range of vortex shedding frequencies, allowing the aerodynamic forces to be utilized most effectively.

The optimization of the stiffness coefficient is achieved through the use of a magnetic resonance tuning system. Magnetic interaction between the structural elements alters the effective stiffness of the system, allowing the resonance frequency to be dynamically adapted to changing wind conditions. This is particularly important for operation under variable wind speeds, as traditional

structures with fixed stiffness have a narrow range of operating frequencies.

2.4 Geometric Parameters and Aerodynamic Efficiency

The shape and dimensions of the structure have a significant impact on vortex shedding characteristics and energy transfer efficiency. To ensure stable vortex formation, the structure should have an optimized diameter-to-height ratio. According to studies on the aerodynamics of cylindrical structures, the optimal length-to-diameter ratio L/D for vortex-induced vibrations is 10–15, which allows for stable and orderly vortex shedding along the entire height of the structure.

In addition, the increase in wind speed gradient with height, described by the Hellmann exponential law,

$$V(h) = V(h_{ref}) \cdot \left(\frac{h}{h_{ref}} \right)^\alpha$$

where $V(h)$ is the wind speed at height h , $V(h_{ref})$ is the wind speed at the reference height, and α is the surface roughness coefficient. This requires accounting for changes in vortex shedding frequency along the mast. To achieve maximum efficiency, a method of varying the structure's diameter along its height can be used, which helps synchronize vortex shedding and increase energy generation efficiency. The vortex shedding frequency directly depends on the airflow velocity and the characteristic length of the structure, so changing the diameter along the height allows stabilization of vortex shedding processes and alignment with the system's mechanical oscillations. This approach promotes more efficient wind energy extraction and reduces aerodynamic losses, thereby increasing the overall efficiency of the generator.

An additional means of improving system efficiency is the use of helical grooves (Fig. 3) along the entire length of the mast. The surface roughness coefficient, denoted as α , plays an important role in shaping the airflow around the structure. The introduction of helical grooves allows adjustment of this coefficient, enhancing the aerodynamic characteristics.

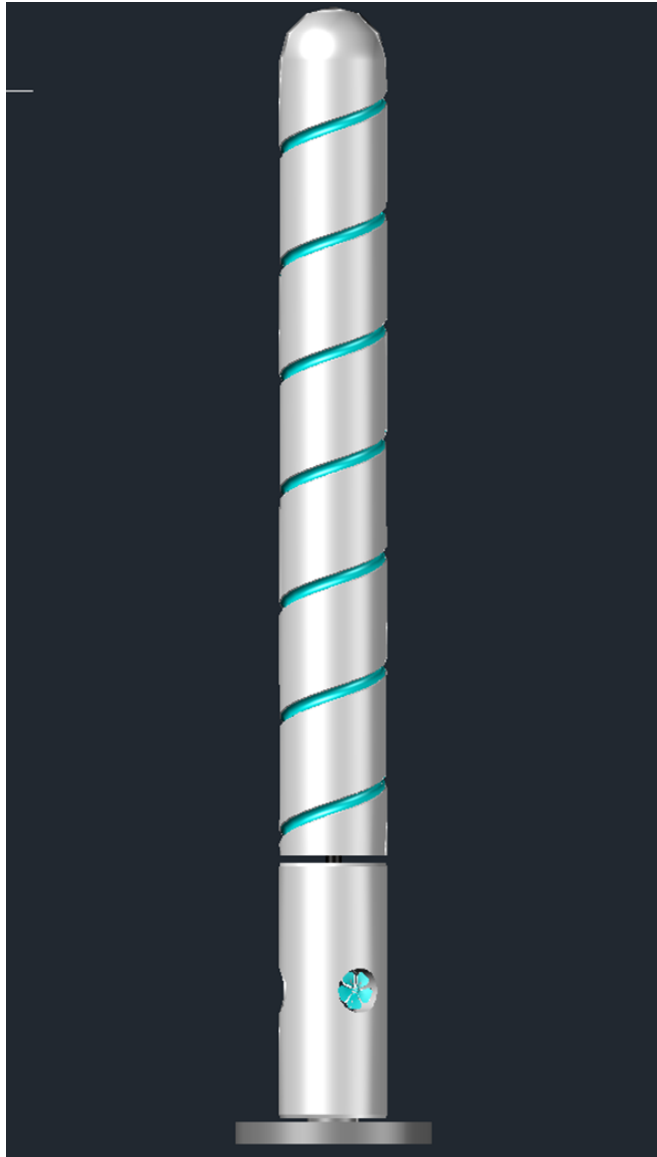


Fig. 3. Helical Grooves on the Wind Turbine Body

The surface roughness coefficient determines the nature of airflow around the structure and affects the boundary layer structure.

Under standard conditions, smooth cylindrical structures exhibit relatively high drag, which can lead to unstable vortex shedding and the occurrence of undesirable vibrations.

The use of helical grooves allows optimization of the airflow interaction with the structure, ensuring controlled vortex formation and stabilization of vortex shedding.

The grooves direct the flow in such a way that chaotic pressure fluctuations, which can cause unstable vibrations of the structure, are reduced. This contributes to lower aerodynamic drag and decreased turbulent losses, which in turn enhances the efficiency of energy transfer to the generation system. This effect is particularly important for a structure utilizing vortex-induced oscillations, as the stability of vortex shedding directly affects the amplitude and frequency of mechanical oscillations transmitted to the internal part of the system.

Numerical experiments show that, with an optimal selection of helical groove parameters, the value of the coefficient α can be reduced by 10–20%, leading to improved interaction of the structure with the airflow and reduced aerodynamic losses.

The optimal inclination angle of the helical grooves relative to the structure's axis is 20–30 degrees, and their depth should be within 1–3% of the mast diameter. Deeper grooves can increase local flow vortices, resulting in additional aerodynamic losses, while grooves that are too shallow have minimal impact on vortex shedding characteristics.

Thus, the combination of varying the structure's diameter along its height and the use of helical grooves allows a significant increase in the system's aerodynamic efficiency, providing stable vortex shedding and improved mechanical energy transfer. These measures contribute to a higher generator efficiency, reduced energy losses, and an expanded range of operating wind speeds, making the structure more reliable and adaptable to changing environmental conditions.

2.5 Resonance Tuning System

The vortex shedding frequency is proportional to the wind speed, which is not constant. However, the range of wind speeds at which the structure resonates is

quite narrow, since the natural frequency of the structure is fixed. To increase the number of operational hours per year, it is necessary to expand this range of effective wind speeds. One strategy for achieving this is the use of a magnetic resonance tuning system.

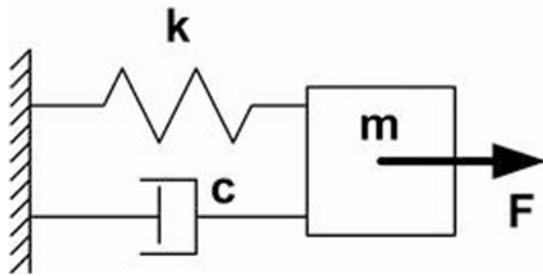
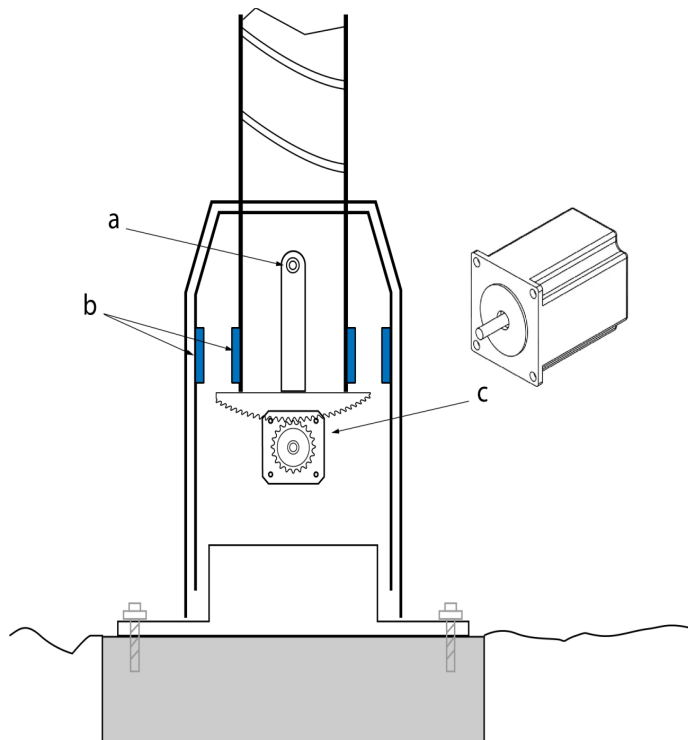


Fig. 4. Diagram of a damped harmonic oscillator

The operating principle of such a system can be represented using a damped harmonic oscillator (Fig. 4), where a mass m oscillates under the action of a force F and is connected to a spring with a damper having a

damping coefficient c . The addition of a pair of permanent magnets allows the stiffness of the system to be adjusted. The magnetic repulsion between them behaves like a spring with a variable stiffness coefficient k .



The generator design (Fig. 5) includes several key components that ensure the efficient operation of the system. The shaft (a) is a bearing mechanism that divides the structure into two parts, L_1 and L_2 . It functions to transmit the oscillations generated by vortex wind flows throughout the system,

Fig. 5. Diagram of the Operating Principle of providing stable motion of the
Magnetic Dampers structure.

The stepper motor (*c*) is installed within the device housing and connected to a rack-and-pinion mechanism. Oscillations of the L_2 part are transmitted through a gear attached to its end and converted into electrical energy with high efficiency.

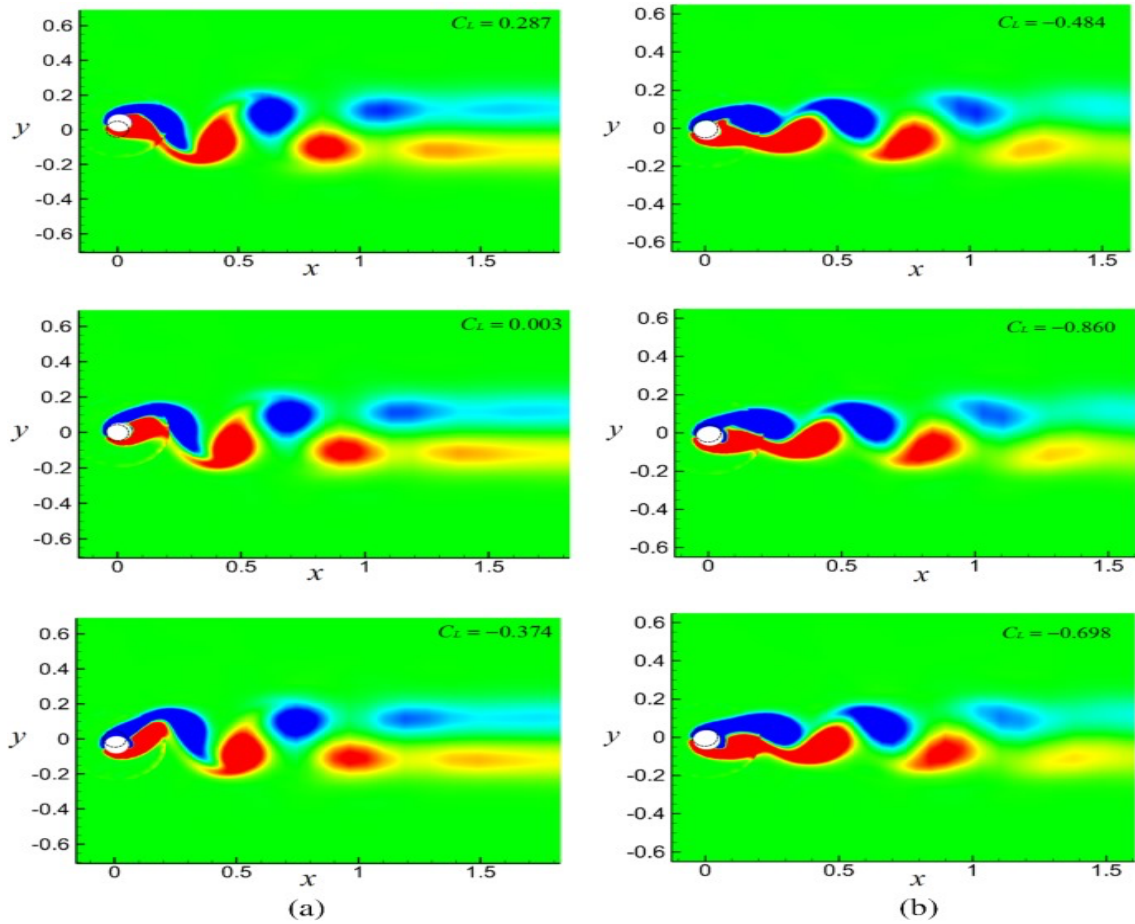
Task 3. Assemble a Ventrix prototype and conduct tests to determine its operating parameters.

3.1 Experimental Data and Modeling

To study vortex-induced oscillations in the generator structure, CFD simulations were conducted to model the interaction of airflow with the cylindrical mast. The plots (Figs. 6–12) illustrate how vortices form around the structure and how the lift coefficient changes at different stages of the oscillation cycle. The color scale represents the pressure distribution: red and blue regions indicate areas of high and low pressure, respectively. The simulation results show that vortices periodically detach from the surface of the structure, generating transverse forces that induce oscillations. This process forms the fundamental operating principle of the generator.

At the initial stages of vortex formation, the first vortices appear and initiate structural oscillations. Subsequently, the pressure difference increases, intensifying the vibrations. At certain moments, the pressure distribution becomes symmetric, temporarily reducing the oscillation amplitude; however, vortex shedding continues. At later stages, the vortices exert maximum influence, causing a change in the direction of the structure's oscillations. In the final stages, vortex shedding stabilizes, and the system enters a steady operating regime.

The analysis of the CFD simulations confirms that the structure efficiently converts vortex-induced oscillations into mechanical energy. The vortex shedding process remains predictable, enabling synchronization of the structure's mechanical oscillations with the wind flow. This ensures stable system operation, reduces energy losses, and allows the generator to function under a wide range of wind speeds. Thus, the modeling results confirm the feasibility of the proposed technology and demonstrate its potential for further optimization.



Figs. 6–12. Vortex shedding contours during the motion cycle of a cylindrical VIV generator ((a) with helical grooves, (b) without).

3.2 Experimental Procedure

To verify the hypothesis regarding the possibility of inducing vortex-induced vibrations (VIV) in the Ventrix structure and to quantitatively evaluate its operating parameters, controlled laboratory tests of the initial prototype were conducted. All measurements were performed indoors on a laboratory test bench providing a regulated airflow, specifically using a guide tube (airflow channel) with adjustable flow velocity. This approach made it possible to obtain reproducible data sets at fixed wind speeds.



Fig. 13. Experimental procedure

During the experiment, it was observed that the structure indeed exhibits oscillations in a direction perpendicular to the airflow vector, which fully corresponds to theoretical expectations. At fixed wind speeds, the following parameters were obtained (Appendix 1).

The analysis of the dependencies revealed three characteristic regimes:

- (a) linear growth of oscillation amplitude at low wind speeds;
- (b) a pronounced resonance peak when the vortex shedding frequency becomes comparable to the natural frequency of the structure;
- (c) a saturation / partial damping zone at higher speeds, where the amplitude either stabilizes or begins to decrease due to nonlinear effects and aerodynamic losses.

Resonance was observed as a local maximum in oscillation amplitude accompanied by an increase in the energy transferred to the generator. This confirms the theoretical model according to which efficient energy harvesting is

achieved when the vortex shedding frequency (determined by the Strouhal number and flow velocity) coincides with the natural frequency of the oscillatory system.

The output power graphs demonstrate a clear correlation with oscillation amplitude: maximum output is observed near resonance, while outside the resonant band the power decreases.

Based on the observations, an oscillation amplitude of 14 cm at a measured frequency of 5 Hz was recorded. These parameters confirm the effectiveness of vortex-induced oscillations and their potential for further use in the process of electrical energy generation.

3.3 Calculation of Potential Energy Output

Electric power generation in the proposed system is based on converting mechanical oscillations into electric current through energy transfer to a stepper motor. The potential power that can be extracted from the mechanical oscillations of the structure is determined using the kinetic energy equation:

$$E_k = \frac{1}{2}mv^2$$

where m is the mass of the oscillating part of the structure, and v is the maximum velocity of the oscillatory motion.

The maximum oscillation velocity is determined from the expression for harmonic oscillations:

$$v_{max} = 2\pi fA$$

where A is the oscillation amplitude and f is the frequency. Substituting the numerical values, we obtain:

$$v_{max} = 2\pi \times 5 \times 0.14 \approx 4.4 \text{ м/с}$$

If it is assumed that the mass of the oscillating part of the structure is 2 kg, then the kinetic energy stored at the peak of motion will be:

$$E = \frac{1}{2} \times 2 \times (4.4)^2 \approx 19.4 \text{ Дж}$$

At a frequency of 5 Hz, the oscillatory system completes 5 cycles per second, resulting in an average power of:

$$P = E \times f = 19.4 \times 5 \approx 97 \text{ Вт}$$

These calculations show that even at the test prototype stage, the structure is potentially capable of generating approximately 97 W of power, which is a significant figure for the initial phase of research. Laboratory tests of the initial prototype confirmed the feasibility of the VIV generation concept on a controlled test bench and allowed identification of the resonance band where maximum output is achieved. These results formed the basis for designing the next modification — V2 with a 360° shaft and an active radial magnet control system, aimed at expanding the operating range and automatically maintaining the optimal amplitude.

Task 4. Propose Ways to Improve the Ventrix Prototype

4.1 Overall Goal of Modernization

Analysis of the experimental data revealed a narrow resonance band and the sensitivity of the initial single-axis design to transitions into excessive amplitude modes during sudden increases in wind speed.

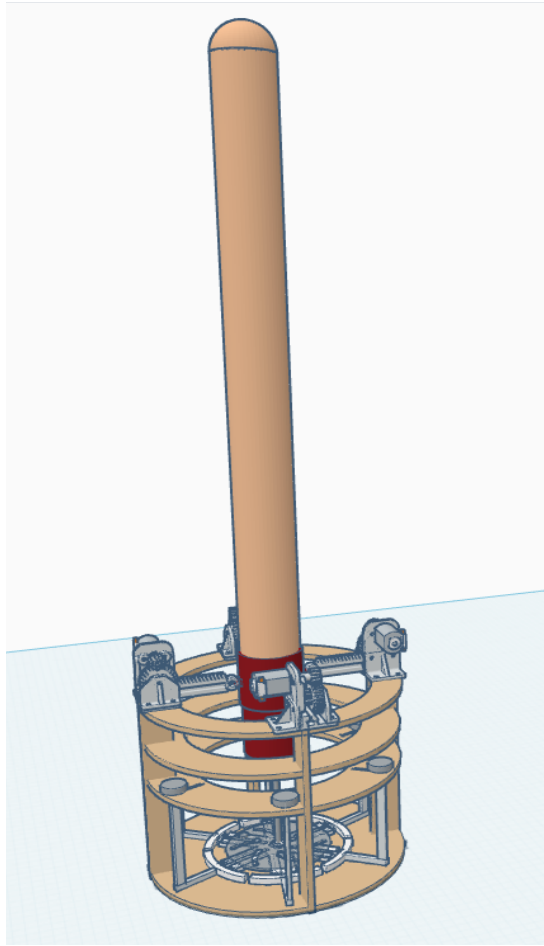
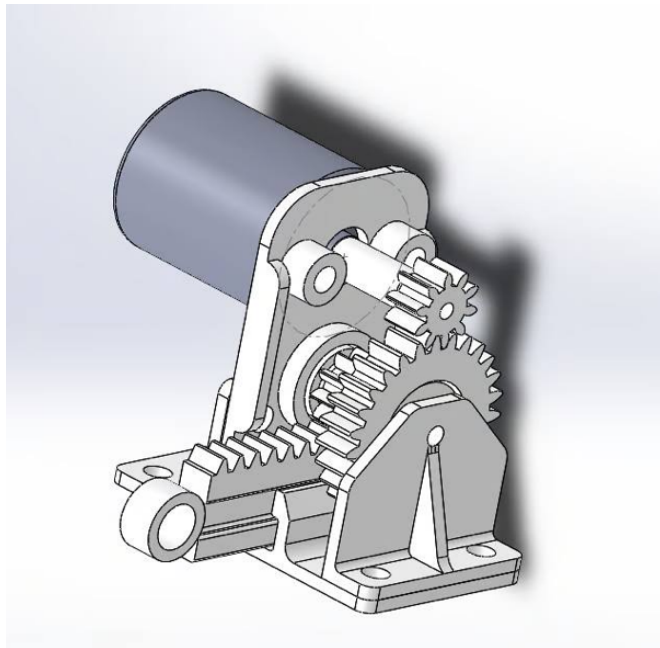
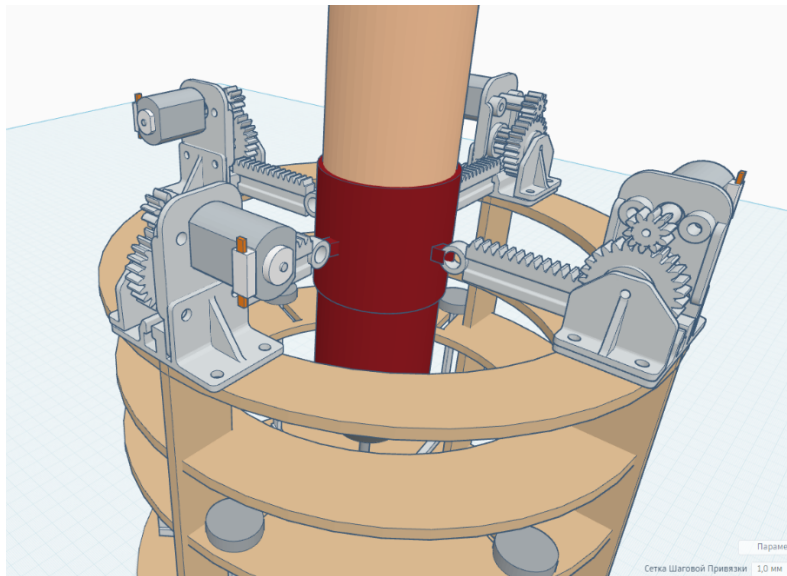


Fig. 14. 3D model of the upgraded design

The goal of the modernization is to expand the range of operating wind speeds, ensure automatic maintenance of amplitude within allowable limits, and increase the structural reliability during strong gusts of wind. This is achieved through a two-level modification: (1) Transition from a single-axis oscillation arrangement to a design capable of rotating around a vertical axis by 360° ; (2) Implementation of an active radial resonance tuning system, realized through an “expanding pulley” mechanism, providing synchronous radial movement of six adjustment magnets.

4.2 Principle of the 360° Shaft

The 360° shaft allows for independent orientation of the elongated mast relative to the incoming flow vector and distributes the excited vortices around the circular perimeter, reducing the likelihood of local overloads and enabling a multi-quadrant energy harvesting scheme (generators positioned around the shaft



Figs. 15, 16. 3D model of the system for 360-degree energy harvesting

4.3 Design of the Magnetic Radial Control System (Concept and Purpose)

The principle of magnetic resonance tuning is based on changing the effective “stiffness” of the oscillatory system by varying the magnetic repulsion/attraction

capture oscillations regardless of the flow direction). The practical implementation is based on a bearing assembly with a rotary flange at the lower part of the housing, allowing free or driven rotation of the structure around the vertical axis.

between the adjustment magnets and the fixed central magnet. Amplitude stabilization is achieved by altering the radial position of six adjustment magnets around the elongated mast: moving the magnets closer to the central element increases the repulsive magnetic force (increasing effective stiffness) → reducing the amplitude; moving the magnets away decreases stiffness → amplitude increases.

4.4 “Expanding Pulley” Mechanism – Structural and Kinematic Description

For synchronous radial movement of the six magnets in the Ventrrix V2, an “expanding pulley” type mechanism is used. A central disk with radial slots is connected to pins that drive six petals, each of which is linked to its magnet via a guide tube with longitudinal slots (3 cm in length). When the disk rotates, the petals simultaneously move apart or together, ensuring the magnets move toward or away from the center.

The disk rotation is driven by a gear transmission connected to an electric motor, which is activated by a controller signal.

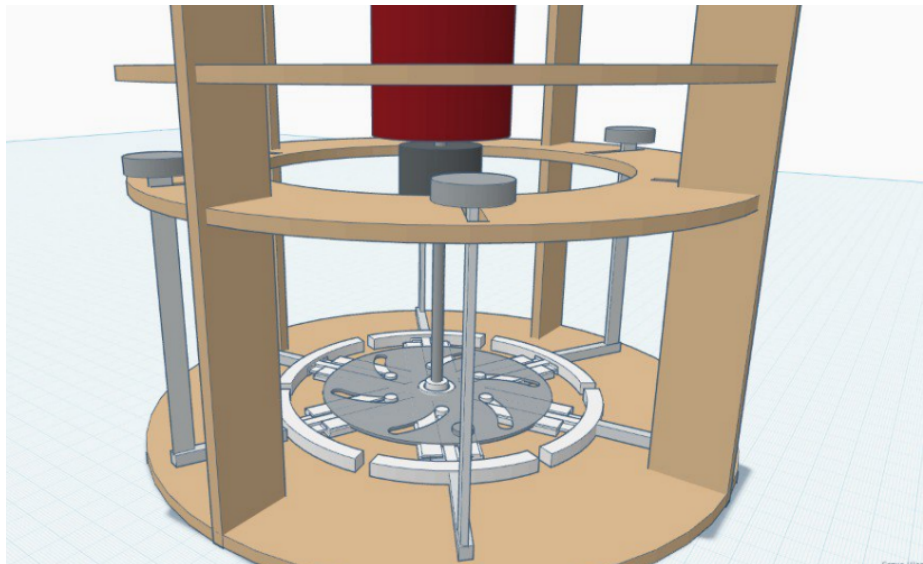


Fig. 17. 3D model of the radial magnet movement mechanism for amplitude control

The controller receives data from four mini-generators that measure oscillation amplitude and output power. When the amplitude exceeds a threshold, the magnets move closer (stiffness increases, amplitude decreases); when it drops, the magnets move away (sensitivity and output increase).

In the event of a power failure, the system locks the magnets in a safe position. In the 3D model, the gear transmission and electric drive are not shown, but they are included in the real prototype design.

During the project, an advanced Ventrix mechanism was thoroughly developed, incorporating a 360° shaft and a radial magnet adjustment system based on the expanding pulley. The proposed design enables amplitude control of oscillations, stability during strong winds, and continuous 24/7 generator operation. The mechanism has been fully engineered, and in the coming months, the author plans to begin its practical implementation in the updated prototype.

Conclusion

In this study, the concept of the bladeless Ventrix wind turbine, based on the principle of vortex-induced vibrations (VIV), was developed and experimentally validated. The analysis demonstrated that this mechanism can efficiently convert wind kinetic energy into electrical energy, offering an alternative to traditional turbines, which are characterized by high cost, noise, and environmental risks.

The experimental Ventrix prototype confirmed the possibility of stable energy generation, reaching approximately 97 W under laboratory conditions. The results showed that the vortex-induced vibration technology is effective even at low and variable wind speeds, making it particularly promising for the climatic conditions of the Zhambyl region.

Of particular significance is the upgraded Ventrix version, equipped with a radial magnet adjustment system and a 360-degree shaft. Thanks to the built-in adaptive amplitude control mechanism, the device can automatically adjust to changes in wind flow and prevent structural overload. This makes Ventrix the first prototype in the world capable of continuous 24/7 operation, whereas all existing turbines require shutdowns under strong wind loads.

Thus, Ventrix represents an innovative and sustainable solution to the energy deficit problem in the Kordai region. Its implementation will enable local electricity generation, reduce dependence on external supplies, and demonstrate a pathway toward environmentally safe and technologically advanced energy in Kazakhstan.

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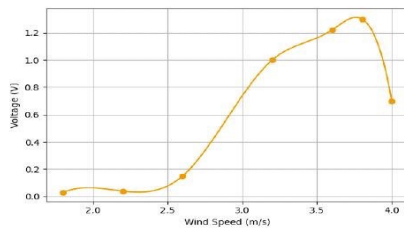
Appendix 1:



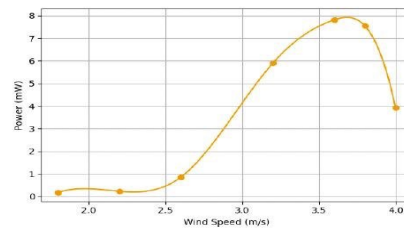
Fig. 1. Visualization of vortex-induced excitations around a cylindrical object

Appendix 2:

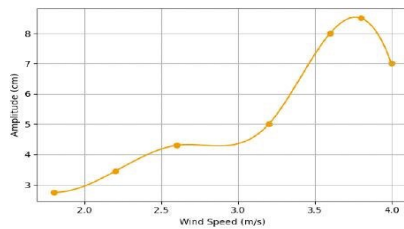
No	Wind Speed (m/s)	Voltage (V)	Current (mA)	Power (mW)	Amplitude (cm)
1	1.80	0.03	5.55	0.17	2.75
2	2.20	0.04	5.60	0.22	3.45
3	2.60	0.15	5.67	0.85	4.30
4	3.20	1.00	5.90	5.90	5.00
5	3.60	1.22	6.40	7.81	8.00
6	3.80	1.30	5.80	7.54	8.50
7	4.00	0.70	5.60	3.92	7.00



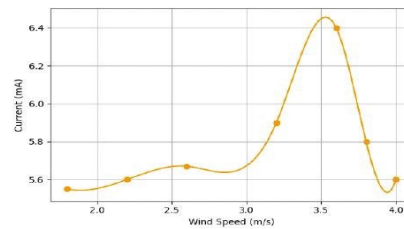
Voltage vs. Wind Speed



Power vs. Wind Speed



Amplitude vs. Wind Speed



Current vs. Wind Speed

Table 1. Experimental Output Data

Figs. 1–4. Dependencies of power, voltage, amplitude, and electrical output on airflow velocity